

# Haitian Liang (梁海天)

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## Education

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2023.09 – Present      Nankai University      B.Sc. in Mathematical Sciences

Major GPA: 3.81/4.0, Top 25% (Overall GPA: 3.64/4.0)

Courses: Mathematical Analysis, Advanced Algebra, Data Mining, Operations Research, Optimization Methods, Combinatorics, Real Analysis, Numerical Analysis, Bioinformatics

English Proficiency: Passed CET-4 & CET-6; capable of basic communication and competent in academic writing.

### Awards:

- Provincial First Prize, National High School Mathematics Competition
- National Third Prize, Chinese Mathematical Olympiad (CMO)
- Provincial First Prize, National High School Informatics Competition
- Meritorious Winner (First Prize, Top 4%), The Mathematical Contest in Modeling (MCM)
- Third Prize, The Chinese Mathematics Competitions for College Students (CMC)

## Research Experiences

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### Experience in Nankai University, 2024-2026 (Mentor: Prof. Shengquan Chen)

#### (1) A systematic benchmarking framework and dual-view optimization strategy for single-cell DNA methylation imputation (Core independent research project completed over 2 years)

Responsible for experimental design, benchmarking, and development of a dual-view (both data and method) optimization strategy derived from in-depth analytical insights.

Manuscript being submitted to a **Top journal** (IF = 7.7, **First author**), is now after Major Revision.

(Code and Introduction is available at <https://github.com/BioX-NKU/scMethyImputeBenchmarks>)

The manuscript was presented at the *3rd Undergraduate Academic Forum in Mathematics in Wuhan University (2026.3)* and the *Forum on AI-Enabled Health Big Data Analysis (2025.11)*.

#### (2) A User-Friendly Database for Single-Cell Chromatin Accessibility Research

Contributed to the collection, curation and annotation of 81 public datasets to build a well-annotated database for single-cell chromatin accessibility sequencing (scCAS).

Manuscript accepted in April 2025 by a **JCR Q1 journal** (IF = 4.6; DOI: 10.1007/s11704-025-41390-5).

### Visiting Student in Westlake University, 2025 (Mentor: Chao Tang, Academician of CAS and Xinya Zhang)

Visited Westlake University; be involved in the Summer Training Program at Peking University Center for Quantitative Biology (PKU-CQB), gaining interdisciplinary experience in quantitative scientific research;

Participated in ongoing research: AI-based interdisciplinary scientific research.

### Summer Research in HKUST, 2026 (Mentor: Prof. Wei You)

Since July, 2026, I will participate in the **official summer research program** of the Department of Industrial Engineering and Decision Analytics (**IEDA**) in HKUST on decision analytics and optimization.

## Internship Experiences

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2025.02 – 2025.06      ByteDance Data Services Department

Participated in work related to the Lean4 language and data annotation for mathematics competition problems.

2023 – 2025      Harbin Normal University High School

Involved in teaching mathematics competition courses with strong presentation skills.

Served as the chief editor for the number theory section of a mathematics competition textbook (published in 2025), and cultivated excellent expression and logical thinking abilities.

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